



SEA-BIRD
SCIENTIFIC

SBE37-SMP-ODO MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16243
Instrument Firmware Version: 2.13.0
Zero Conductivity Frequency: 2784.56
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10755354	2000m(2000 dBar)
RS232	Oxygen	SBE 63	63-1753	7000m

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



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SENSOR SERIAL NUMBER: 16243
CALIBRATION DATE: 01-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

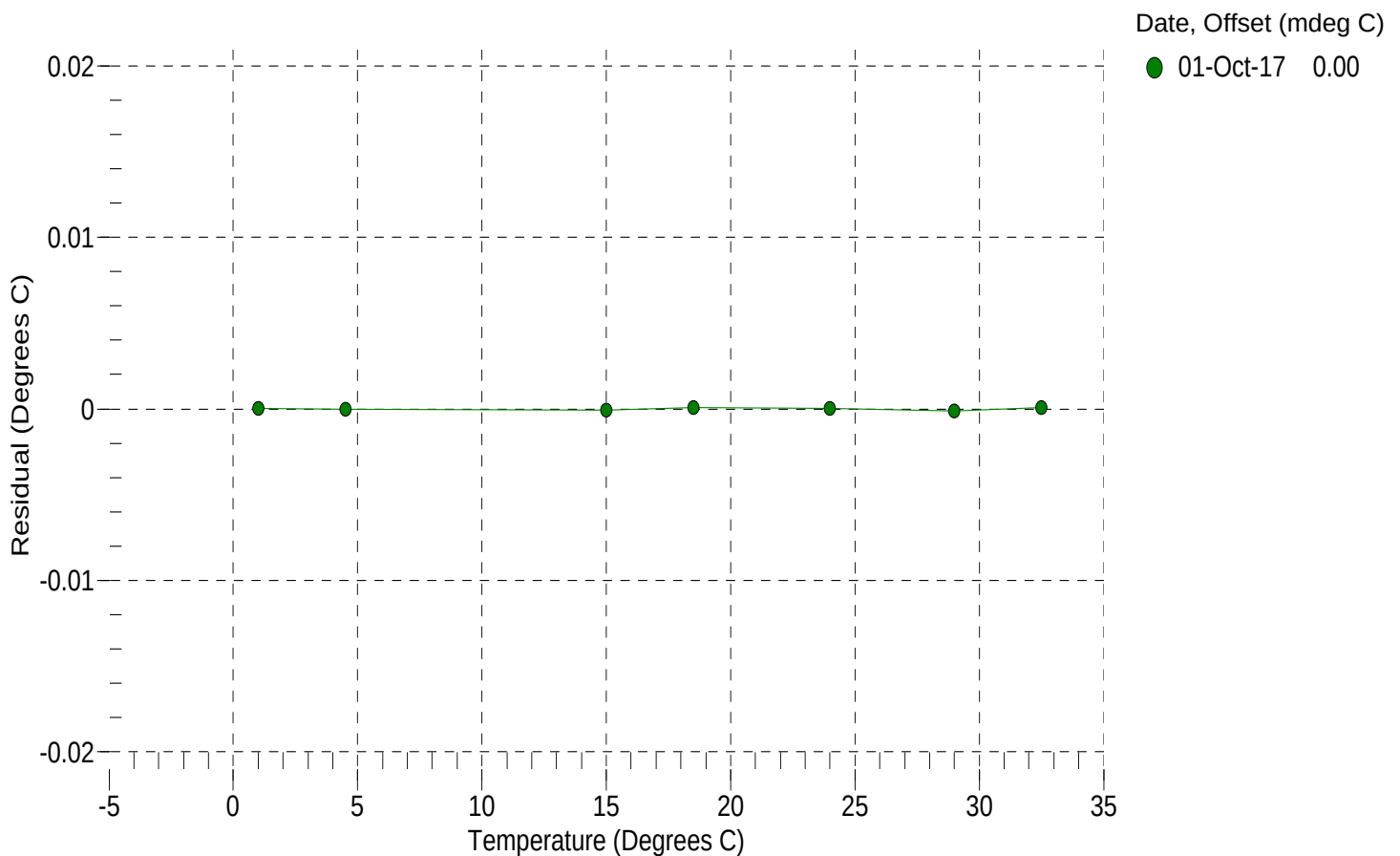
a0 = -1.947572e-004
a1 = 3.235149e-004
a2 = -5.635965e-006
a3 = 2.333318e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	572888.9	1.0000	0.0000
4.5000	490606.2	4.5000	-0.0000
15.0000	314177.8	14.9999	-0.0001
18.5000	272485.9	18.5001	0.0001
24.0000	219162.3	24.0000	0.0000
29.0000	180894.8	28.9999	-0.0001
32.5000	158681.3	32.5001	0.0001

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -9.887249e-001
h = 1.275678e-001
i = -8.039014e-005
j = 2.218102e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = -1.1970e-007

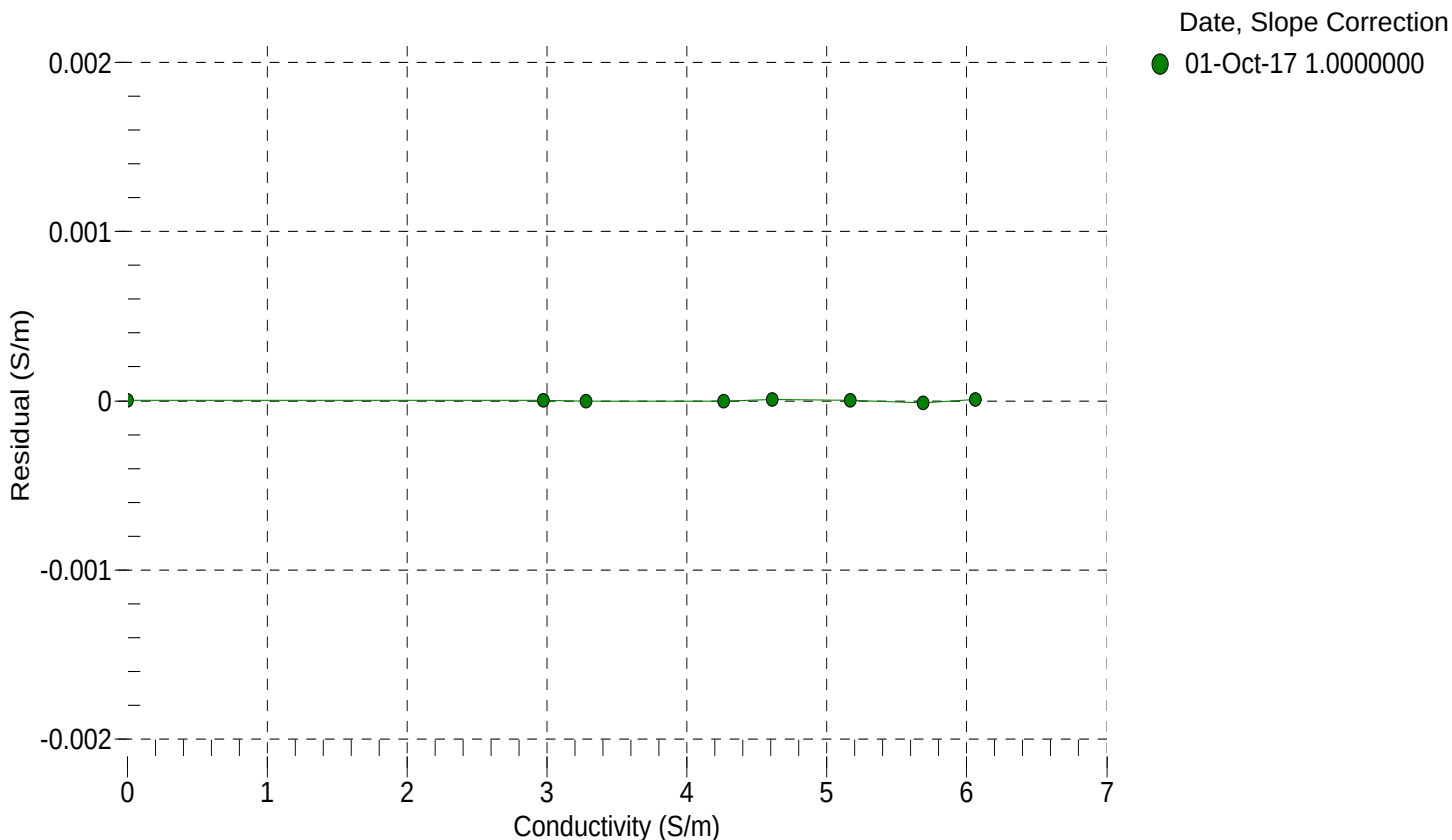
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2784.56	0.00000	0.00000
1.0000	34.7935	2.97420	5568.39	2.97420	0.00000
4.5000	34.7740	3.28113	5779.21	3.28112	-0.00000
15.0000	34.7327	4.26244	6406.11	4.26244	-0.00000
18.5000	34.7244	4.60749	6612.17	4.60750	0.00001
24.0000	34.7157	5.16531	6932.03	5.16532	0.00000
29.0000	34.7109	5.68700	7218.04	5.68698	-0.00001
32.5000	34.7083	6.05928	7415.23	6.05929	0.00001

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16243
CALIBRATION DATE: 29-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10755354

COEFFICIENTS:

PA0 = 3.944372e-001
PA1 = 8.994230e-003
PA2 = -1.511347e-010
PTEMPA0 = -8.275429e+001
PTEMPA1 = 4.652482e-002
PTEMPA2 = -1.453723e-007

PTCA0 = 5.243163e+005
PTCA1 = 3.908602e+000
PTCA2 = -8.684016e-002
PTCB0 = 2.521812e+001
PTCB1 = 2.506266e-005
PTCB2 = 0.000000e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.59	525940.4	2282.5	14.61	0.00	32.50	2497	526005.95
591.74	590183.3	2289.5	591.75	0.00	29.00	2421	526012.54
1168.89	654560.7	2290.3	1168.86	-0.00	24.00	2311	526017.56
1746.05	719090.2	2291.1	1746.07	0.00	18.50	2191	526014.93
2323.18	783751.0	2292.0	2323.19	0.00	15.00	2115	526008.90
2900.25	848540.8	2292.3	2900.20	-0.00	4.50	1887	525986.88
2323.17	783759.9	2291.9	2323.27	0.00	1.00	1811	525976.57
1746.16	719092.9	2291.9	1746.09	-0.00	TEMPERATURE (°C) SPAN (mV)		
1168.93	654569.6	2291.5	1168.94	0.00			
591.68	590172.2	2291.1	591.65	-0.00			
14.59	525938.2	2292.1	14.59	0.00	-4.90	25.22	25.22
					35.00	25.22	

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

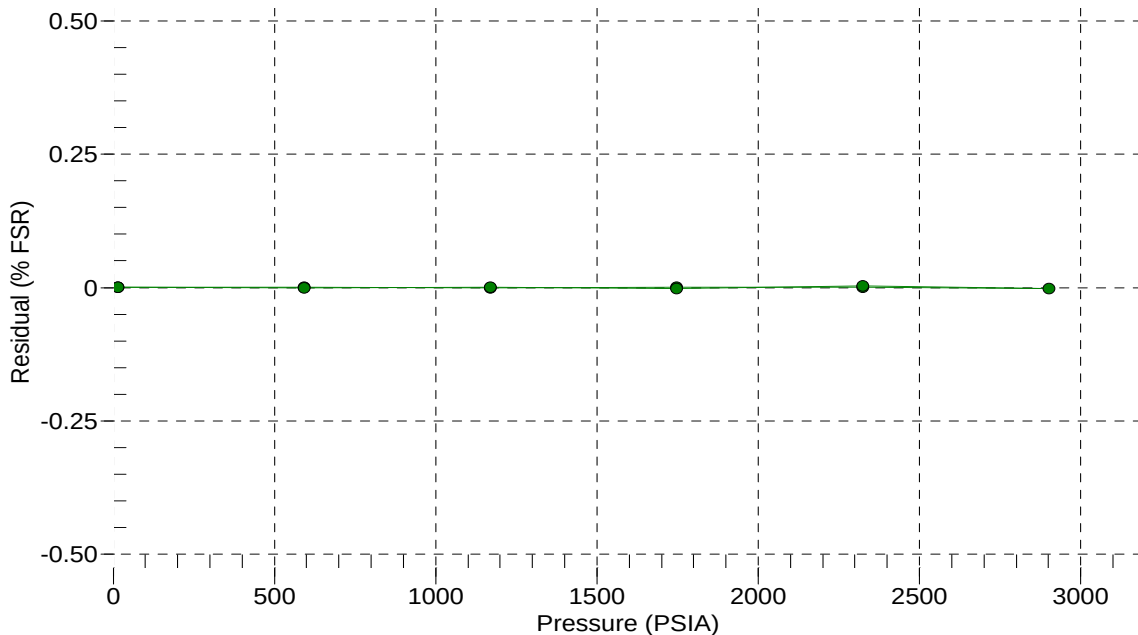
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 29-Sep-17 -0.00





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Pressure Test Certificate

Test Date: **2017-09-28**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16243**

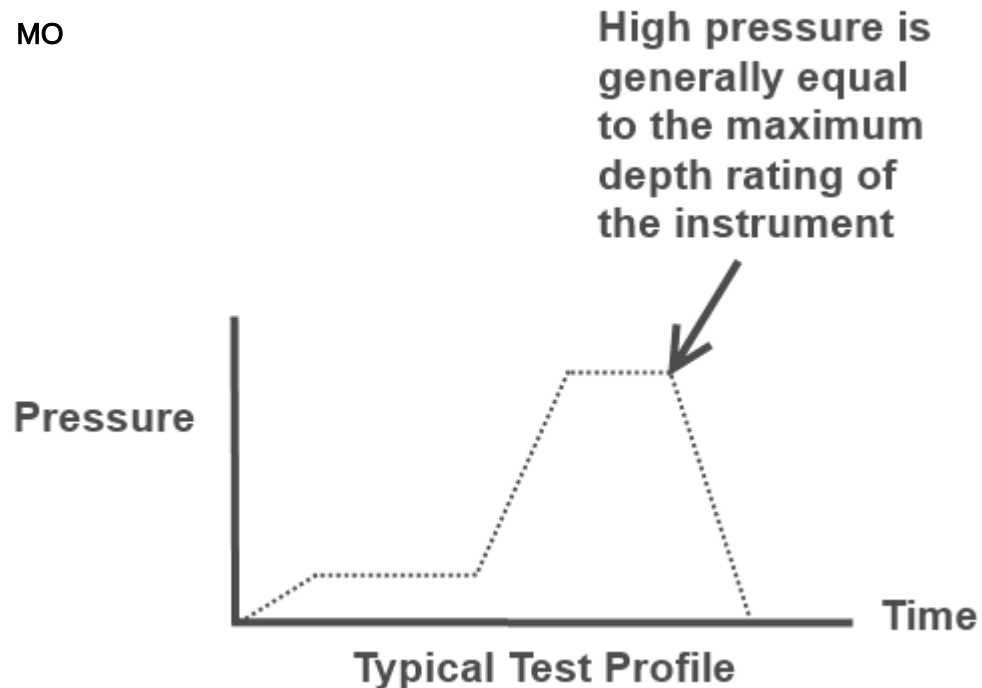
Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **MO**





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SENSOR SERIAL NUMBER: 1753
CALIBRATION DATE: 19-Sep-17

SBE 63 OXYGEN TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

TA0 = 6.755729e-004 TA2 = -4.474674e-008

TA1 = 2.584945e-004 TA3 = 1.251896e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT(V)	INST TEMP (° C)	RESIDUAL (° C)
2.0000	1.12107	1.9998	-0.00020
2.0000	1.12106	2.0001	0.00011
2.0001	1.12105	2.0004	0.00032
2.0002	1.12106	2.0001	-0.00009
5.9999	0.99677	5.9998	-0.00014
6.0000	0.99676	6.0001	0.00009
6.0000	0.99677	5.9998	-0.00024
6.0002	0.99676	6.0001	-0.00011
12.0001	0.83072	12.0007	0.00058
12.0001	0.83073	12.0003	0.00019
12.0002	0.83073	12.0003	0.00009
12.0002	0.83074	11.9999	-0.00030
20.0000	0.64682	19.9999	-0.00007
20.0001	0.64682	19.9999	-0.00017
20.0001	0.64682	19.9999	-0.00017
20.0001	0.64682	19.9999	-0.00017
26.0001	0.53464	25.9999	-0.00018
26.0001	0.53463	26.0005	0.00041
26.0001	0.53463	26.0005	0.00041
26.0001	0.53464	25.9999	-0.00018
29.9999	0.47071	29.9999	0.00004
29.9999	0.47071	29.9999	0.00004
30.0000	0.47071	29.9999	-0.00006
30.0001	0.47071	29.9999	-0.00016

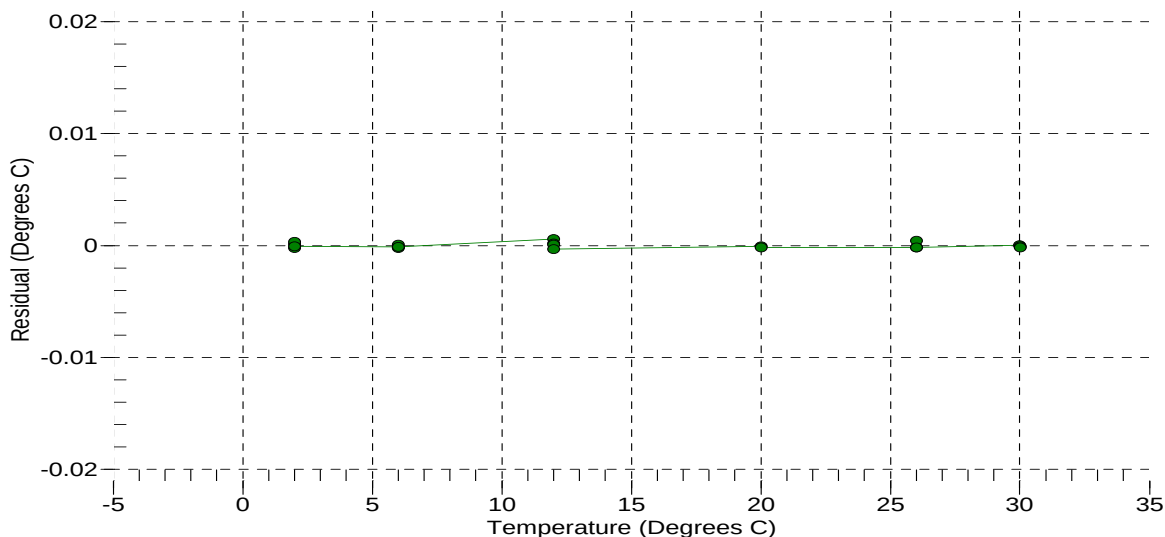
V = Instrument Output (Volts)

$L = \ln(100000 * V / (3.3 - V))$

Temperature ITS-90 (°C) = $1 / (TA0 + (TA1 * L) + (TA2 * L^2) + (TA3 * L^3)) - 273.15$

Residual (°C) = instrument temperature - bath temperature

Date, Offset (mdeg C)
● 19-Sep-17 -0.00





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SBE 63 OXYGEN CALIBRATION DATA

COEFFICIENTS:

A0 = 1.0513e+000 B0 = -2.4892e-001 C0 = 1.0886e-001 E = 1.1000e-002
A1 = -1.5000e-003 B1 = 1.5977e+000 C1 = 4.6178e-003
A2 = 3.6504e-001 C2 = 6.4788e-005

BATH OXYGEN (ml/l)	BATH TEMPERATURE (° C)	BATH SALINITY (PSU)	INSTRUMENT OUTPUT (µsec)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
0.745	30.00	0.00	30.89	0.745	0.000
0.773	26.00	0.00	31.55	0.774	0.000
0.820	20.00	0.00	32.58	0.823	0.003
0.895	12.00	0.00	34.05	0.894	-0.000
1.018	6.00	0.00	34.87	1.018	-0.000
1.113	2.00	0.00	35.46	1.112	-0.000
2.225	30.00	0.00	22.73	2.223	-0.002
2.349	26.00	0.00	23.31	2.348	-0.001
2.498	20.00	0.00	24.43	2.504	0.006
3.006	12.00	0.00	25.37	3.005	-0.001
3.410	6.00	0.00	26.35	3.408	-0.002
3.690	30.00	0.00	18.80	3.684	-0.006
3.773	2.00	0.00	26.96	3.772	-0.001
3.926	26.00	0.00	19.26	3.926	0.000
4.334	20.00	0.00	20.01	4.343	0.009
5.095	12.00	0.00	21.06	5.091	-0.004
5.285	30.00	0.00	16.24	5.281	-0.004
5.652	26.00	0.00	16.61	5.653	0.001
5.841	6.00	0.00	21.89	5.839	-0.002
6.261	20.00	0.00	17.27	6.269	0.008
6.464	2.00	0.00	22.47	6.465	0.001
7.313	12.00	0.00	18.24	7.309	-0.004
8.359	6.00	0.00	19.03	8.356	-0.003
8.861	2.00	0.00	19.90	8.865	0.004

T = temperature (°C) , P = pressure (dbar), U = Instrument output (µsec)

S_{corr} (salinity correction function) = 1.0 for calibration in DI water

See the user manual for more information on S_{corr} calculation

$V = U / 39.457071$

Oxygen (ml/l) = $\{((A0 + A1 \cdot T + A2 \cdot V^2) / (B0 + B1 \cdot V) - 1.0) / (C0 + C1 \cdot T + C2 \cdot T^2)\} \cdot S_{corr} \cdot \exp(E \cdot P / (T + 273.15))$

Residual (ml/l) = instrument oxygen - bath oxygen

Date, Slope Correction

● 19-Sep-17 1.0000

